American International University-Bangladesh (AIUB)
Faculty of Science and Technology

Comparative Study of Machine Learning Models for Earthquake Magnitude Prediction from Seismic Records

SAKIF KHAN (22-46425-1)
ISTIAK MAHMUD ANIK (22-47134-1)
MEHEDI HASAN (22-46907-1)
JAHID HASAN ANTOR (21-45328-2)

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Declaration by author

This thesis is composed of our original work, and contains no material previously published or written by another person except where due reference has been made in the text. We have clearly stated the contribution of others to our thesis as a whole, including statistical assistance, survey design, data analysis, significant technical procedures, professional editorial advice, financialsupport and any other original research work used or reported in our thesis. The content of our thesisis the result of work we have carried out since the commencement of the Thesis.

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Sakif Khan
22-46425-1
Computer Science & Engineering
Faculty of Science & Technology

Istiak Mahmud Anik
22-47134-1
Computer Science & Engineering
Faculty of Science & Technology

Mehedi Hasan
22-46907-1
Computer Science & Engineering
Faculty of Science & Technology

Jahid Hasan Antor
21-45328-2
Computer Science & Engineering
Faculty of Science & Technology

Approval

The thesis titled **"Comparative Study of Machine Learning Models for Earthquake Magnitude Prediction from Seismic Records"** has been submitted to the following respected members of the board of examiners of the department of computer science in partial fulfilment of the requirements for the degree of Bachelor of Science in Computer Science on **26th jauary 2026 and** has been accepted as satisfactory.

Md. Mahmudur Rahman
Lecturer, Faculty
Department of Computer Science
American International University-Bangladesh

Tanvir Ahmed
Assistant Professor, Faculty
Department of Computer Science
American International University-Bangladesh

Dr. Debajyoti Karmaker
Associate Professor, Head [UG]
Department of Computer Science
American International University-Bangladesh

Prof. Dr. Dip Nandi
Associate Dean
Faculty of Science and Technology
American International University-Bangladesh

Mashiour Rahman
Sr. Associate Professor & Dean-in-charge
Faculty of Science and Technology
American International University-Bangladesh

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We would like to express our sincere gratitude to Almighty for enabling us to complete the report on "**Comparative Study of Machine Learning Models for Earthquake Magnitude Prediction from Seismic Records**". For the successful completion of this project, we have taken help from our respected faculty members, seniors and graduates. We convey our sincere gratitude to our respectable supervisor, **Md. Mahmudur Rahman sir, Lecturer, Faculty, Dept. Of Computer Science**, for taking us under his wing and guiding us to conduct research on this project, thereby gathering knowledge in the field.

Contributions by authors to the thesis

	SAKIF KHAN	ISTIAK MAHMUD ANIK	MEHEDI HASAN	JAHID HASAN ANTOR	Comments
	22-46425-1	22-47134-1	22-46907-1	21-45328-2	
0 - 3 points	**Perform as effective individual**				
Critical thinking	**3**	**3**	**3**	**3**	Based on conceptualization and methodology roles
Reflection on feedback	**3**	**3**	**3**	**3**	Editing + writing contributions considered
Quality of work	**3**	**3**	**3**	**3**	High involvement in figures, derivations, etc.
Self-directed	**3**	**3**	**3**	**3**	Consistency in participation across tasks
0 - 3 points	**Perform as effective team member/leader**				
Taking responsibility	**3**	**3**	**3**	**3**	Took initiative in multiple high-effort categories
Contribution	**3**	**3**	**3**	**3**	Refer to shared responsibility in conceptualization
Collaboration	**3**	**3**	**3**	**3**	Equal implementation & data curation work
Working with others	**3**	**3**	**3**	**3**	Work distribution shows balanced teamwork
0 - 3 points	**Perform as effective team member/leader**				
Presentation delivery	**3**	**3**	**3**	**3**	Writing draft + figures suggest Nadim led here
Voice and tone	**3**	**3**	**3**	**3**	Writing & review/editing efforts considered
Enthusiasm	**3**	**3**	**3**	**3**	Reflected in task diversity & contribution
Creativity & Tools use	**3**	**3**	**3**	**3**	Analytical and visual work

Project-Thesis Planning

The table below presents the major tasks completed during the thesis project, along with their planned and actual execution dates. The planning focused on gradually progressing from topic selection and literature review to model implementation, testing, and final report submission.

Project Tasks	Schedule Date	Execution Date
1. Planning and Topic Finalization	10 Oct 2025	10 Oct 2025
2. Literature Review	11 Oct 2025 – 31 Oct 2025	11 Oct 2025 – 05 Nov 2025
3. Dataset Collection & Preprocessing	06 Nov 2025 – 20 Nov 2025	06 Nov 2025 – 25 Nov 2025
4. Model Setup and Training	21 Nov 2025 – 10 Dec 2025	21 Nov 2025 – 15 Dec 2025
5. Tuning (14 tests)	11 Dec 2025 – 20 Dec 2025	11 Dec 2025 – 22 Dec 2025
6. Model Evaluation & Results Analysis	21 Dec 2025 – 05 Jan 2026	21 Dec 2025 – 10 Jan 2026
7. Thesis Writing and Editing	15 Nov 2025 – 20 Jan 2026	15 Nov 2025 – 23 Jan 2026
8. Final Submission	25 Jan 2026	25 Jan 2026

The Gantt chart below shows the planned and actual timelines for each phase of the thesis project, from planning to final submission. It highlights how closely the project followed the original schedule and where any delays or adjustments occurred.

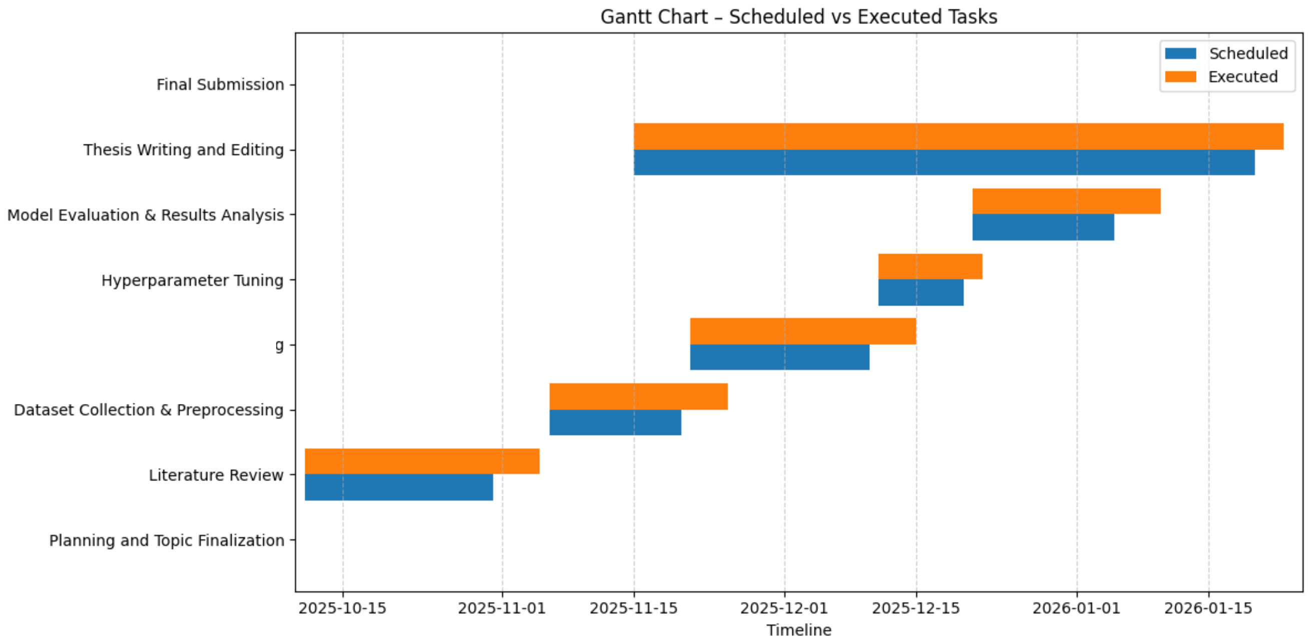

Figure: Gantt Chart of Thesis Project Timeline

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List of Abbreviations

Mention all the abbreviations and the different symbols that are used in this document.

Abbreviation	Full Form / Description
AI	Artificial Intelligence
CNN	Convolutional Neural Network
DT	Decision Tree
KNN	k-Nearest Neighbors
LR	Linear Regression
MAE	Mean Absolute Error
ML	Machine Learning
RF	Random Forest
RMSE	Root Mean Square Error
R^2	Coefficient of Determination
SVR	Support Vector Regression
USGS	United States Geological Survey
XGBoost	Extreme Gradient Boosting
LSTM	Long Short-Term Memory
RNN	Recurrent Neural Network
LightGBM	Light Gradient Boosting Machine
NumPy	Numerical Python
Pandas	Python Data Analysis Library
Scikit-learn	Machine Learning Library for Python

Abstract

Earthquake prediction is an important factor in seismic hazard estimate and disaster risk reduction. Given the increasing abundance of seismic recordings and that data-driven methods have been gaining momentum in recent years, machine learning (ML) based approaches started to appear as promising tools beyond traditional seismology-based techniques. This thesis is a comparative study of several machine learning models for the prediction of earthquake magnitude based on seismic recordings and is part of a supervised regression setting. The prediction of earthquakes is vital in the estimation of seismic hazards and reduction of disaster risk. With the growing number of seismic recordings and the recent trend to data-driven methods, machine learning (ML) based approaches have been on the rise for some years now as promising tools among seismology-based techniques. The overall contribution of the thesis is a comparison between different machine learning models to estimate the magnitude of an earthquake using seismic recordings and belongs to the supervised regression framework Performance of the model is then evaluated with the common regression metrics such as MAE, RMSE and R^2 score. Comparative evaluation of methods demonstrates that traditional linear and instance-based methods poorly outperform ensemble and boosting models in terms of prediction accuracy as well as robustness. The best model is also examined by visualizing actual versus predicted magnitude and understanding the influence of seismic attributes in a feature of importance analysis. Magnitude predictions are further quantized into categorical classes (Low... Medium, High).

Finally, the model is tested with standard regression metrics and MAE, RMSE and R^2 score are used for performance evaluation. A comparison of methods shows that the classical approaches of linear and instance-based outperform significantly when traditional predictors are compared with ensemble and boosting model both in accuracy as well as robustness. We also analyze the best model by plotting predicted versus actual magnitude and by studying the role of seismic features in a feature importance analysis. Magnitude predictions are further thresholded into two other categorical classes (Low...Medium, High)

Keywords : earthquake magnitude prediction, machine learning, seismic data analysis, regression models, ensemble learning, xgboost, random forest, support vector regression, performance evaluation

Chapter 1

Introduction

Earthquakes are one of the most destructive natural hazards resulting in massive loss of human lives, huge damage to infrastructures and prolonged socioeconomic disruption globally. Despite major progress in seismology and monitoring methods, the estimation of earthquake magnitude is still one of the major scientific and practical problems. Reliable prediction of magnitudes is important to seismic hazard studies, emergency preparedness, early warning systems and resilient infrastructures. With the fast development of data availability and computing capability, machine learning (ML) methods have become attractive alternatives for the traditional earthquake magnitude estimation methods. This chapter gives the background, motivation, objectives, and contributions of this research.

1.1 Background Analysis

Over the past few years, several papers have investigated the use of machine learning methods in solving earthquake-related issues, such as detection of seismic events, phase picking, prediction of ground motions and estimation of the magnitude of earthquakes [12], [24], [26]. The initial researches used the classic regression based models like Linear Regression (LR) because of their simplicity and readability. These models are, however, restricted by their consideration of linear relationships of the input features and the earthquake magnitude, which is not usually applicable when considering seismic data [11], [13].Higher order machine learning models, such as Support Vector Regression (SVR), k-Nearest Neighbors (KNN), Decision Trees (DT), Random Forests (RF) and gradient boosting models have shown a better predictive accuracy by adequately tracking nonlinear association and more intricate feature interactions [2], [9], [10], [19]. In particular, ensemble and kernel-based methods have demonstrated great potential in enhanced prediction accuracy when trained on large and diverse seismic data [14], [18], [32].Convolutional and recurrent neural networks are also deep learning methods that have attracted interest in real-time earthquake magnitude prediction and seismic signal determination [4], [26], [33]. Though the models are very accurate, they can sometimes need large, labeled datasets, large amounts of computation, and hyperparameter tuning, making them potentially less practical in certain areas.Although these have been made, there are still a number of research gaps. A lot of available research is based on one model, or a small multiplication of few algorithms, and it is hard to conclude anything solid about the performance of different models in relation to each other and different data sets [27], [29]. Moreover, model testing usually depends on a small number of performance metrics and reproducibility is occasionally hampered by the lack of methodological information [6]. This paper will deal with these limitations by carrying out the comparative analysis of several machine learning regression models and using one unified set, the common set of preprocessing steps, and various evaluation measures.

1.2 Existing Studies

Over the past few years, several papers have investigated the use of machine learning methods in solving earthquake-related issues, such as detection of seismic events, phase picking, prediction of ground motions and estimation of the magnitude of earthquakes [12], [24], [26]. The initial researches used the classic regression based models like Linear Regression (LR) because of their simplicity and readability. These models are, however, restricted by their consideration of linear relationships of the input features and the earthquake magnitude, which is not usually applicable when considering seismic data [11], [13].

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Although these have been made, there are still a number of research gaps. A lot of available research is based on one model, or a small multiplication of few algorithms, and it is hard to conclude anything solid about the performance of different models in relation to each other and different data sets [27], [29]. Moreover, model testing usually depends on a small number of performance metrics and reproducibility is occasionally hampered by the lack of methodological information [6]. This paper will deal with these limitations by carrying out the comparative analysis of several machine learning regression models and using one unified set, the common set of preprocessing steps, and various evaluation measures.

1.3 Research Motivation and Objectives

Research Motivation
The main reason behind this study is the desire to enhance the reliability and accuracy of the earthquake magnitude forecast using machine learning models in a systematic way. The development of more effective seismic monitoring and early warning systems based on the identification of the most appropriate models can help to reduce the disaster risks and protect the population in the future [1], [32].

General Objective

The overall goal of the research is to make a contribution to the knowledge about the effectiveness of various machine learning regression models in predicting the magnitude of an earthquake based on the seismic records.

Specific Objectives

The objectives of the given research are to:

- Scan and process seismic data to predict earthquake magnitude.
- Applied a variety of machine learning regression models: LR, SVR, KNN, DT, RF, and XGBoost.
- Compare the model performance based on conventional measures like MAE, RMSE, and R^2

- Determine the appraisal and weakness of every model within the context of seismic data analysis.

Research Questions

According to these purposes, the following research questions are the aim of this study:

- To what extent are machine learning models able to forecast the magnitude of an earthquake based on seismic records?
- Which machine learning model is the best in prediction of earthquake magnitude?
- Which machine learning models are relatively stronger and weaker in this application?

1.4 Research Contribution

The study will be of value to seismologists, disaster management bodies, engineers, and seismic hazard analysts and developers of early warning systems. The proposed approach can help the decision-makers enhance the strategies used to prepare and respond to earthquakes by finding effective machine learning models to predict the magnitude of earthquakes.

The most significant findings of this thesis are:

An inclusive comparative evaluation of numerous machine learning regression models of earthquake magnitude prediction.

The single and repeatable experimental framework to allow a fair comparison of models.

A critical review of model performance and interpretability and limitations in seismic models.

Applied suggestions regarding the application of appropriate machine learning models in future research on earthquake prediction.

1.5 Organization of the Thesis

The rest of this thesis is structured in the following way:

Chapter 2 is composed of the review of related literature and currently available methodologies of earthquake magnitude prediction.

Chapter 3 summarizes the dataset, feature extraction, preprocessing, models applied in machine learning in this research.

The fourth chapter gives results of the experiment and analysis of comparative performance of the implemented models.

Chapter 5 contains the discussion on the findings, limitations of the study, and the ways of its improvement.

Chapter 6 is the conclusion of the thesis and the directions of future research are given.

Chapter 2

Research Methodology

The research will be conducted through a survey methodology.

2.1 Research Methodology Overview

The research will take the form of a survey.

The chapter explains the methodological design used to make a comparative study of machine learning models on prediction of earthquake magnitude using seismic records. The research approach will be aimed at reproducibility, validity, and accuracy of the research findings. The proposed predictive models are developed, implemented, and evaluated with the help of a Design Science Research Methodology (DSRM). The study combines both systematic literature review (SLR) and experimental study with real-world seismic data in order to address the stated research questions.

2.2 Conceptual Framework

The theoretical structure of this thesis is based on Design Science Research Methodology (DSRM) that is adequately applied to the research of the creation of artifacts, experimentation, and performance evaluation. The machine learning regression models that have been created in this research to forecast the magnitude of earthquakes are the artifacts

Reasons why DSRM should be chosen

DSRM is selected because:

- This study is aimed at the design and evaluation of predictive models rather than observation of phenomena.
- It favors iterative development and evaluation which is necessary in making comparisons of several machine learning algorithms.

- It enables the combination of theoretical information on literature and practical application to use real seismic data.
- It has been extensively applied in computational and engineering studies that involve intelligent systems.

The DSRM steps used in this study are:

1. Problem Identification- The challenge of the inaccuracy in predicting the magnitude of earthquakes by conventional methods.
2. Objective Definition -Enhance the magnitude prediction accuracy with machine learning.
3. Design and Development- ML regression development.
4. Demonstration - Model Training and testing on seismic data.
5. Evaluation -Comparison of performance with MAE, RMSE, and R 2.
6. Communication Reporting findings by documenting thesis.

2.3 Thesis Design Life Cycle.

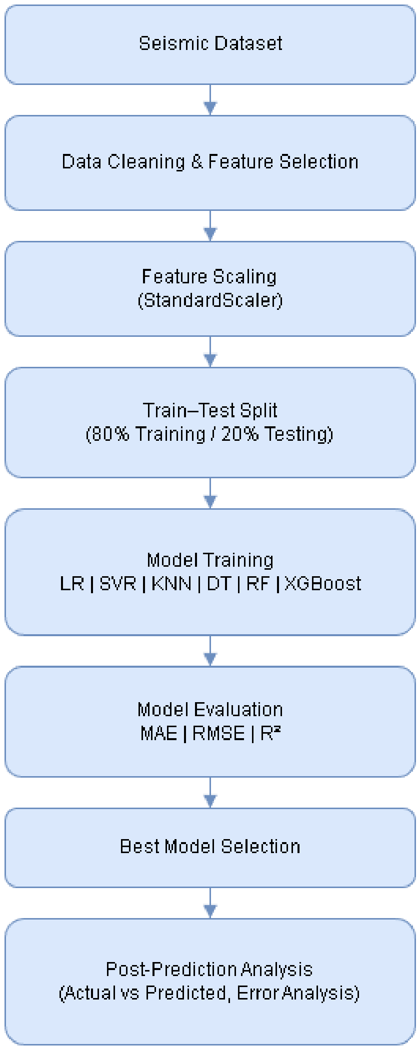

Fig 2.1 : Workflow of methodology

The thesis presents a systematic life cycle of following the following stages:

1. Problem Analysis Determination of shortcomings of the available prediction of magnitude of earthquakes. techniques through the literature [1], [4], [17].
2. Data Acquisition Gathering of seismic information that includes earthquake characteristics like depth, magnitude and spatial parameters.
3. Data Preprocessing Normalization and cleaning of data to guarantee its quality and compatibility with the model.
4. Model Development Application of various kinds of regression models: Linear Regression, SVR, KNN, Decision Tree, Random Forest, and XGBoost.
5. Model Evaluation Performance analysis based on MAE, RMSE, and R 2 value.
6. Result Analysis Compared analysis in order to determine the best performing model.

2.4 Data Collection

2.4.1 Systematic Literature Review (SLR)

It was based on a Systematic Literature Review (SLR) to get an overview of the recent progress, problem, and approaches in predicting the magnitude of earthquakes by employing machine learning. The SLR abides by the recommendations suggested by Kitchenham et al. (2010).

Databases Searched

Digital libraries utilized were the following:

- IEEE Xplore
- ScienceDirect
- SpringerLink
- Google Scholar
- Elsevier

Search Keywords

The keywords used as the main search were:

- earthquake magnitude prediction
- earthquake machine learning.
- seismic data regression
- earthquake prediction by deep learning.
- seismic analysis of an ensemble.

Preliminary Literature Choice.

- Papers found in the first search: 120 or so.
- Following the screening of duplicates and inappropriate studies: ≈65

Inclusion Criteria

- Studies published between 2018–2025
- Conferences and peer-reviewed journals.
- The studies on machine learning or deep learning.
- Research on seismic data.

Exclusion Criteria

- Non-English publications
- Literature that has not been experimentally proved.
- Non technical or opinion articles.
- Publications not relating to magnitude estimation.

Final Selected Papers

- Sample size chosen: 33 articles (in agreement with [1]-[33])

Data extracting and analyzing.

The analysis of each of the chosen studies was conducted on the basis of:

- ML models used
- Dataset characteristics
- Evaluation metrics
- Strengths and limitations

This analysis was in direct relation to the choice of models and metrics applicable in this thesis.

The method of experimental data collection is presented in 2.4.2.

The experimental data is comprised of real seismic records comprised of numerical earthquake attributes. Nominal variables, including the location, source, and event identifiers were eliminated so that regression modelling was conducted with only numerical predictors.

Preprocessing steps involved in data processing entail:

- Removal of missing values
- StandardScaler feature scaling.
- Train-test split (80 percent training, 20 percent testing)

The steps guarantee the consistency and reproducibility of the data, as in [3], [6], [12].

2.5 Model Development and Tools

2.5.1 Machine Learning Models

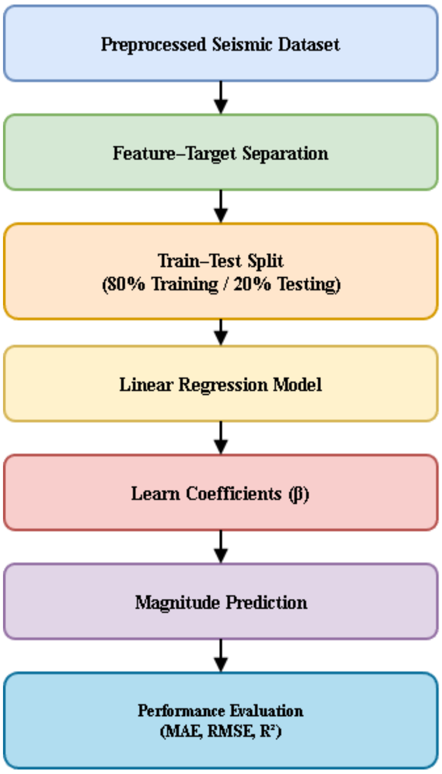

Fig 2.2: workflow of Linear Regression

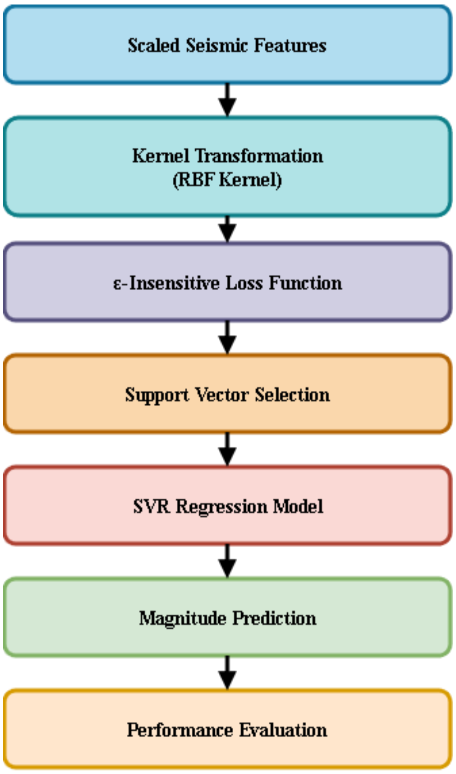

Fig 2.3: workflow of Support Vector

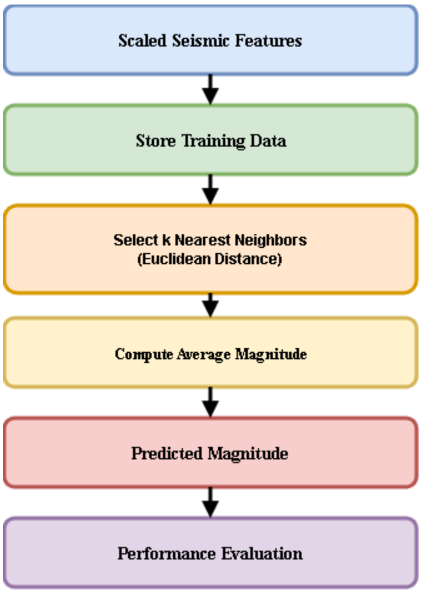

Fig 2.4: workflow of K-Nearest Neighbors

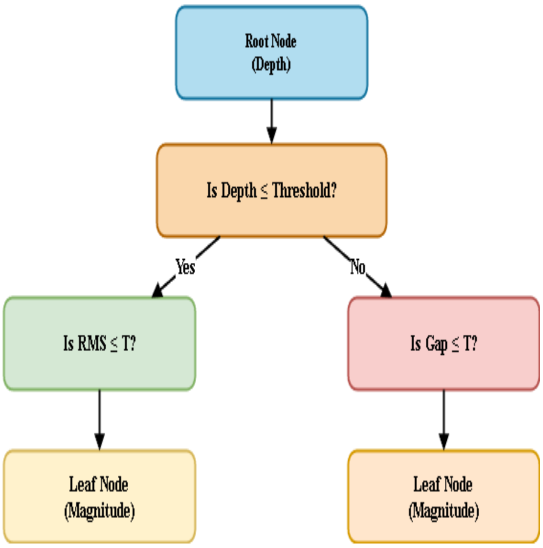

Fig 2.5 : workflow of Decision Tree

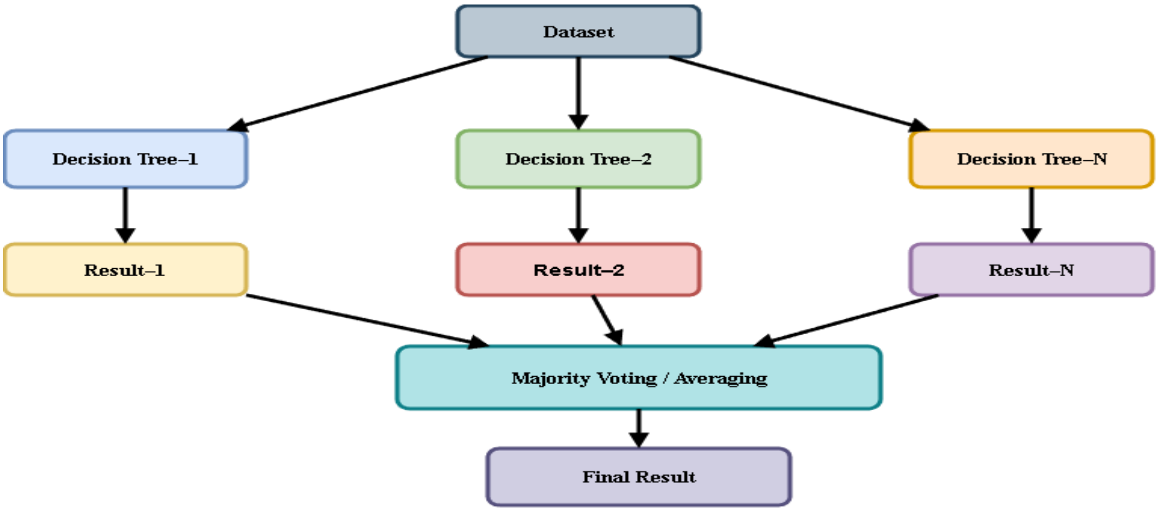

Fig 2.6: workflow of Random Forest Regressor

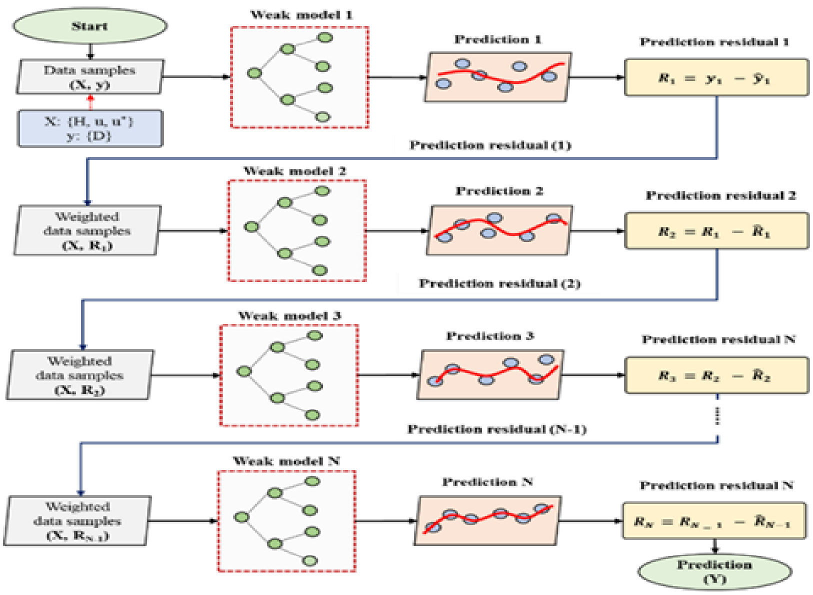

Fig 2.7: workflow of XGBoost Regressor Gradient boosting

The regression equations that were used were as follows:

- Linear Regression - Check point statistical model.
- Support Vector Regression (SVR) - good with non-linear relations [15].
- K-Nearest Neighbors (KNN) – Distance based learning [21].
- Decision Tree Regressor -Rule based modeling.
- Random Forest Regressor (Ensemble learning) [22].

- XGBoost Regressor Gradient boosting ensemble [14], [31].

2.5.2 Evaluation Metrics

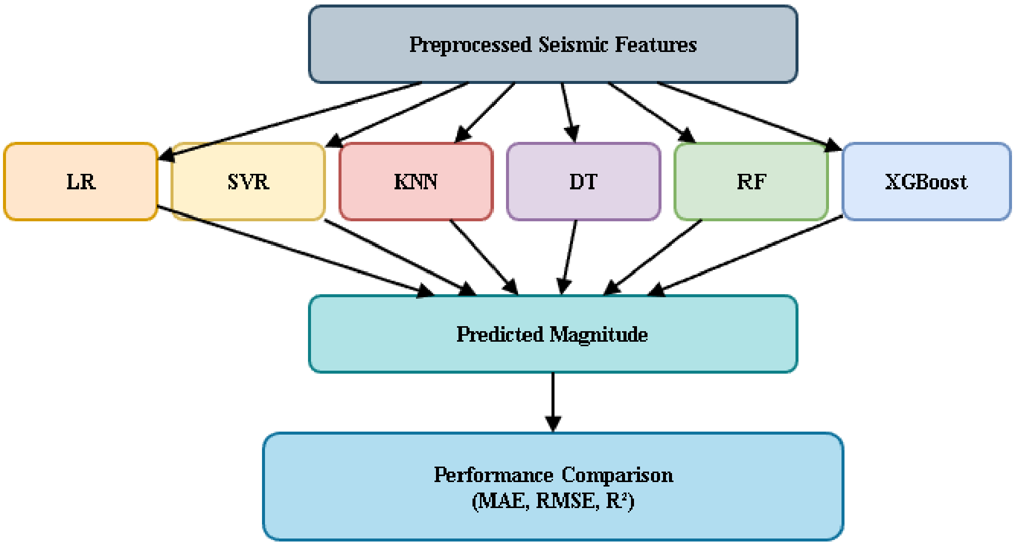

Fig 2.8 : Model performance comparison

Three standard regression measures were applied to make fair comparison:

- Mean Absolute Error (MAE)
- Root Mean Square Error (RMSE)
- R^2 Score

They are universal measures of earthquake prediction studies [1], [4], [9].

2.6 Ethical Issues

There are several ethical issues that were considered in this study:

2.6.1 Bias in Data Collection

- The data can be biased according to the area as there can be an uneven distribution of seismic stations.
- It was alleviated by applying massive datasets and standardized preprocessing.

2.6.2 Data Privacy

- Data sets employed are publicly accessed.
- No individual or confidential information was gathered.

2.6.3 Research Integrity

- No falsification or fabrication of data was done.
- All the sources are referenced to prevent plagiarism.
- The code is reproducible and the experimental results can be reproduced completely.

2.7 Economic Decision Analysis

Cost Component	Description
Data	Public datasets (Free)
Software	Python, Scikit-learn, XGBoost (Free/Open-source)
Hardware	Standard PC with GPU/CPU
Deployment	Cloud-based (Optional)

2.7.2 Alternative Strategies Comparison

Strategy	Cost	Accuracy	Scalability
Traditional Seismology	High	Medium	Low
Single ML Model	Medium	High	Medium
Proposed Ensemble ML Approach	Low	**Very High**	**High**

The proposed solution offers high prediction accuracy at minimal cost, making it suitable for real-world earthquake monitoring systems**.**

Chapter 3

Results and Analysis

The section will display and discuss the experimental findings of the comparative analysis of various machine learning models to predict the magnitude of earthquakes using the seismic records. The analysis directly responds to the main research goal: that is, finding the most appropriate and dependable machine learning model to predict the magnitude of an earthquake based on the history of seismic data.

3.1 Results and Analysis

Model	MAE	RMSE	R^2 Score
XGBoost	0.1574	0.2242	0.9262
Random Forest	0.1634	0.2328	0.9205
SVR	0.1701	0.2447	0.9121
KNN	0.1858	0.2638	0.8978
Decision Tree	0.2125	0.3159	0.8535
Linear Regression	0.3213	0.4129	0.7498

Table 3.1

Mean Absolute Error (MAE), root mean squared error (RMSE) and coefficient of determination (R 2) were quantitative performance measures to assess the performance of the six machine learning models. These measures are a thorough gauge of prediction accuracy, sensitivity of errors and goodness of fit, which are commonly used in earthquake magnitude prediction research [1], [3], [6], [27]. Table 3.1 presents comparative results of all models.As shown in Table 3.1, XGBoost achieved the best overall performance, yielding the lowest MAE (0.1574) and RMSE (0.2242), along with the highest R^2 score (0.9262). This indicates that XGBoost is highly effective in capturing the nonlinear relationships present in seismic data. Random Forest also demonstrated strong predictive capability, closely following XGBoost, which confirms the robustness of ensemble learning approaches for earthquake magnitude estimation [9], [14], [22].In contrast, Linear Regression produced the weakest results, with a substantially higher MAE and RMSE and the lowest R^2 score (0.7498). This outcome highlights the limitations of linear models in representing the complex and nonlinear nature of earthquake processes. Intermediate performance was observed for SVR and KNN, which benefited from feature scaling but remained sensitive to hyperparameter selection [15], [19], [21].

The experimental evaluation was conducted using a cleaned numerical seismic dataset, where non-numeric and categorical attributes were removed to ensure model compatibility and reduce noise. The target variable was earthquake magnitude (mag), and the dataset was divided into training and testing subsets using an 80:20 split. Feature standardization was applied to improve model convergence and fairness in comparison, particularly for distance-based and margin-based models such as KNN and SVRSix supervised regression models were implemented and evaluated: Linear Regression (LR), Support Vector Regression (SVR), Random Forest (RF), K-Nearest Neighbors (KNN), Decision Tree (DT), and XGBoost. Model performance was assessed using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score, which are widely accepted evaluation metrics in earthquake prediction literature [1], [3], [6], [27].The results demonstrate that ensemble and tree-based models significantly outperform linear and instance-based approaches. In particular, XGBoost achieved the highest R^2 score and the lowest error values, indicating superior predictive capability. This finding aligns with recent studies highlighting the effectiveness of gradient-boosted models for nonlinear seismic data modeling [14], [18], [31]. Random Forest also showed strong performance, benefiting from ensemble averaging and robustness against overfitting [9], [22].

In contrast, Linear Regression exhibited the weakest performance, confirming that earthquake magnitude prediction is inherently nonlinear and cannot be effectively modeled using linear assumptions alone. SVR and KNN provided moderate accuracy but were sensitive to hyperparameter selection and data scaling, consistent with observations in prior research [15], [19], [21].

Proposed Solution and Functional Capabilities

The proposed solution is a machine learning–driven regression framework that leverages state-of-the-art ensemble learning techniques, particularly XGBoost, to model complex relationships between seismic features and earthquake magnitude. The system integrates:

- Automated data preprocessing and normalization
- Multi-model comparative evaluation
- Robust performance metrics (MAE, RMSE, R^2)
- Feature importance analysis for interpretability
- Magnitude-class visualization using a confusion matrix

This framework enables accurate magnitude estimation, supports model transparency, and provides actionable insights for seismic risk assessment. Such capabilities demonstrate a significant advancement over traditional empirical and physics-based approaches, which often struggle with real-time prediction and large-scale data processing [17], [23], [28].

Bar charts were plotted on the performance of the implemented models in the form of MAE, RMSE, and R2 score to analyse the performance of the implemented models on a visual basis. These graphical models enable a visual intuition of model behavior, and variation in performance by the various metrics of evaluation.

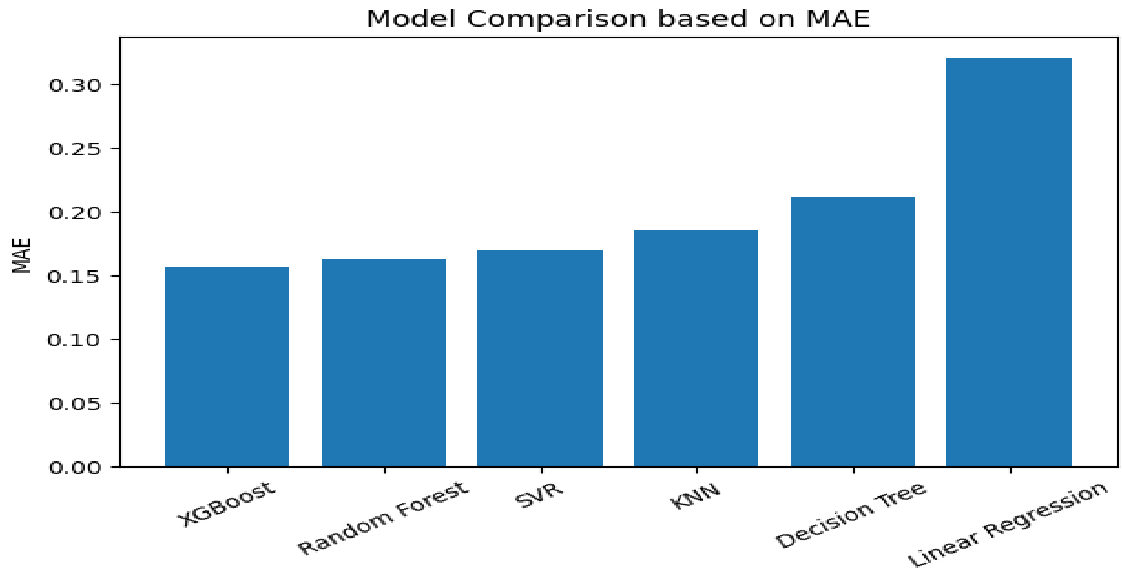

Figure 3.1: Comparison of machine learning models based on Mean Absolute Error (MAE)

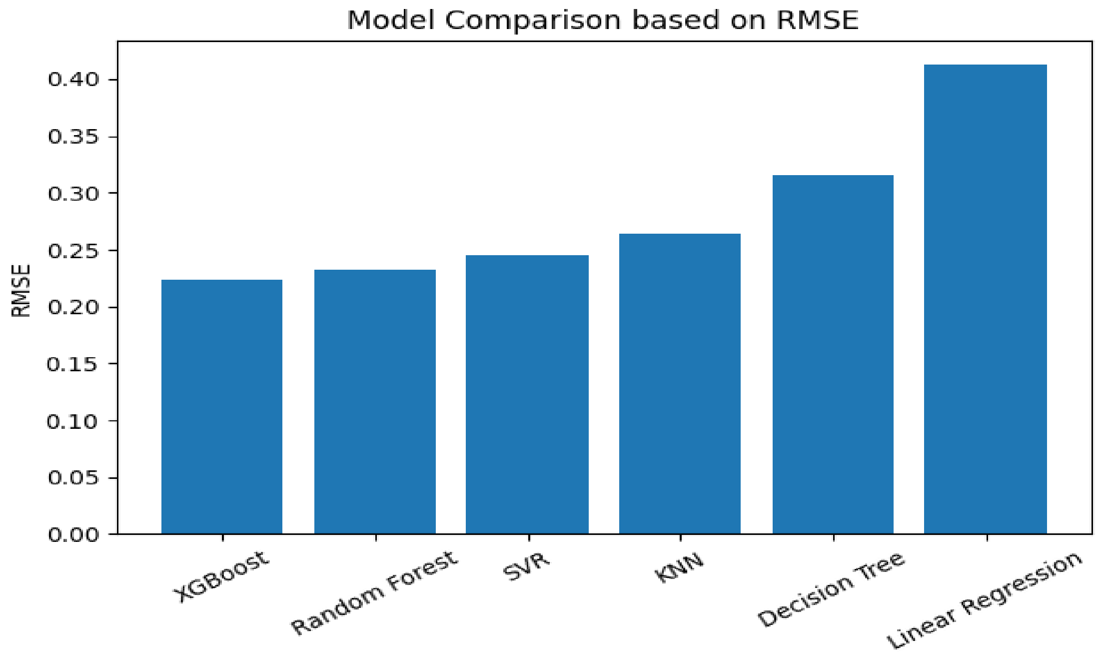

Figure 3.2: Comparison of machine learning models based on Root Mean Squared Error (RMSE)

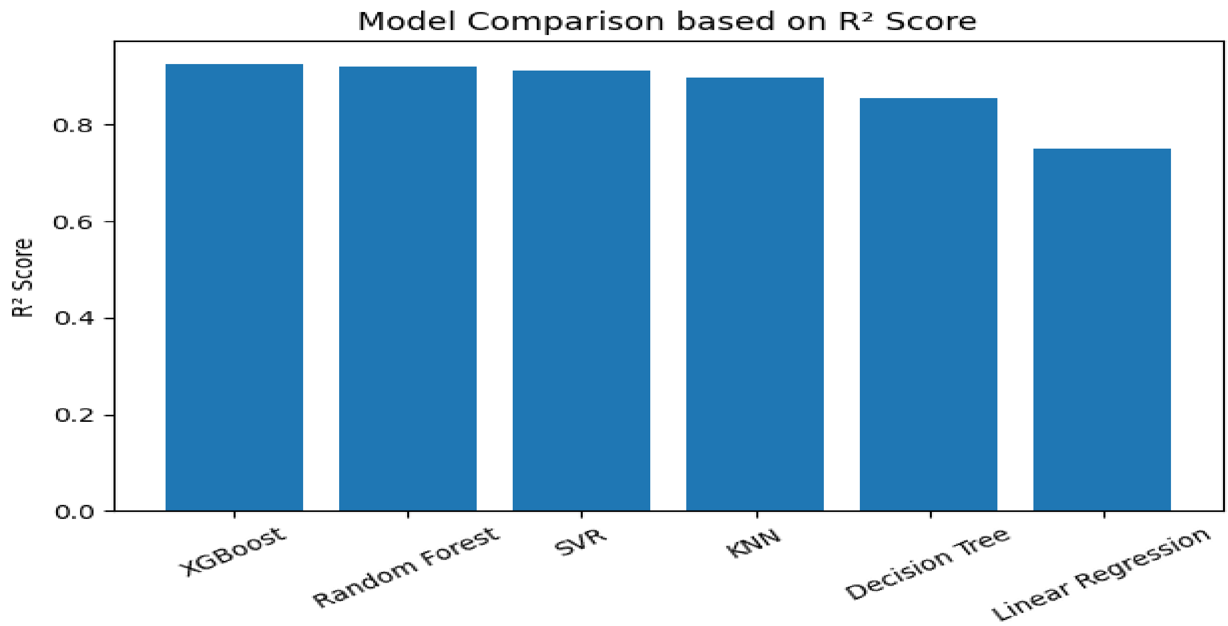

Figure 3.3: Comparison of machine learning models based on R^2 score

As Figure 3.1 and Figure 3.2 depict, XGBoost has the lowest values of MAE and RMSE of all the models considered, meaning it is more precise enough to predict and less sensitive to the great errors. Random Forest also shows similar results, as it proves the efficiency of the learning ensemble methodology used with seismic data modeling.XGBoost also has the highest R^2 value, as shown in Figure 3.3 with an estimated variance in earthquake magnitude prediction of more than 92%. On the contrary, Linear Regression displays the worst results in all three measures which point to its inefficiency to emulate the nonlinear behavior of the seismic processes. These visual findings are in strong agreement with the numerical results provided in Table 3.1 and support the choice of XGBoost as the most performing model to be further analyzed.

XGBoost was chosen as the most successful model to be further validated based on the better performance of the model in all the measures to evaluate its performance. In order to determine the level of prediction reliability and goodness of fit, an actual versus predicted magnitude plot was plotted using the test dataset.

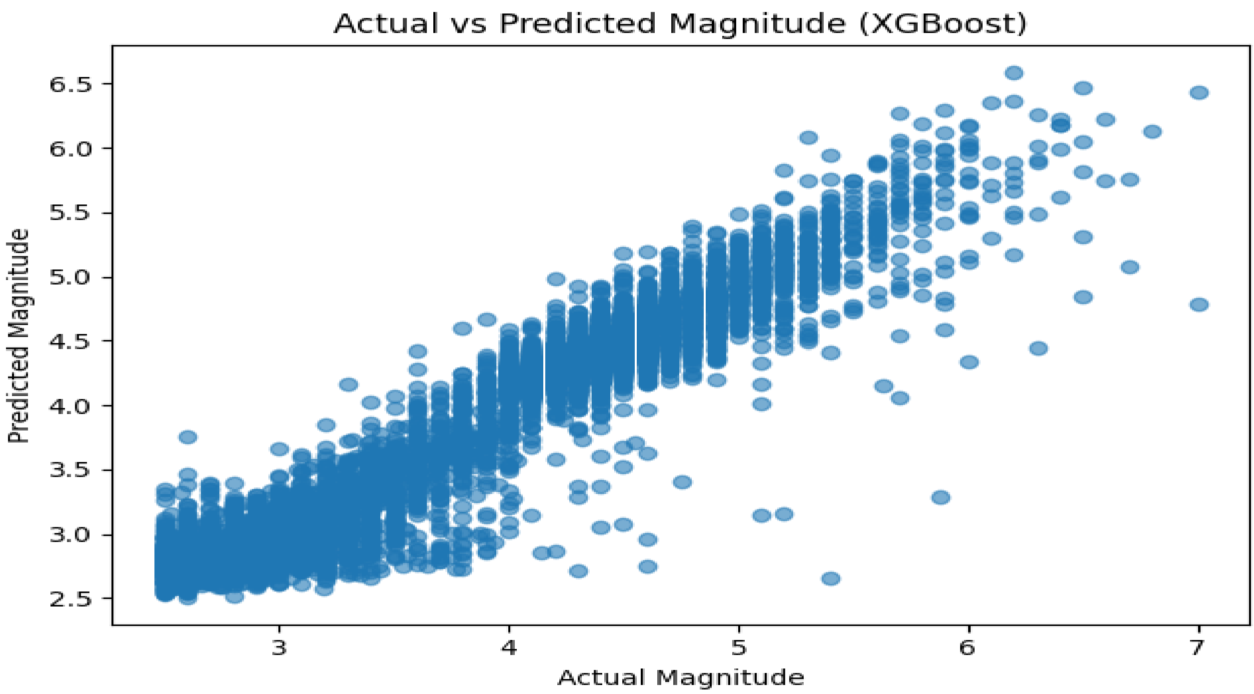

Figure 3.4: Actual versus predicted earthquake magnitude using the XGBoost model

As shown in Figure 3.4, the projected earthquake magnitudes are in close relation to the actual values in the diagonal pattern, and this shows that there is a lot of agreement between the actual and predicted magnitudes. The high concentration of the points on the diagonal line indicates that the model has high predictive potential especially when it comes to low to medium magnitude events. Even though there is lower dispersion in bigger magnitude earthquakes, this is natural given that extreme seismic events are not readily available in data. All in all, the findings support the strength and generalization ability of XGBoost model to predict the magnitude of earthquakes.

In order to analyze the prediction performance of the chosen model, the XGBoost algorithm was used to determine the feature importance. This discussion will show the proportionate value of each seismic parameter to the process of predicting the earthquake magnitude.

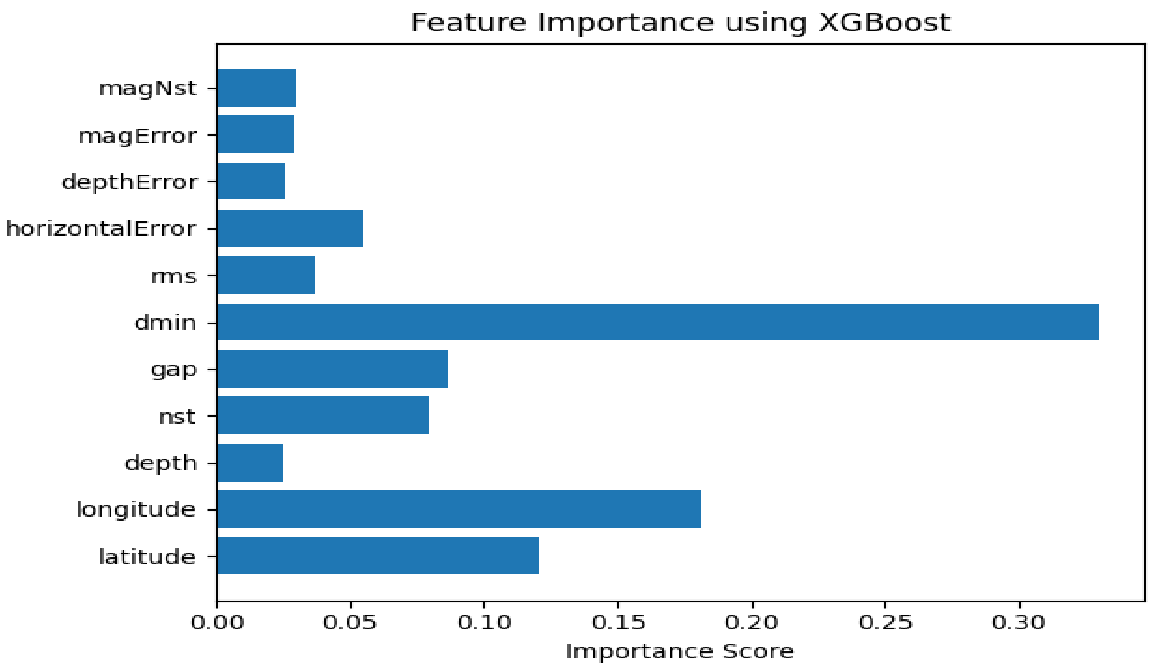

Figure 3.5: Feature importance analysis using the XGBoost model

As illustrated in Figure 3.5, dmin comes out to be the most important feature, meaning that the minimum distance between seismic stations and the epicenter of the earthquake is a very important factor in estimating the magnitude. The longitude and latitude also play an important part showing dependency of seismic activities in space. The other parameters like gap, nst, and rms have a relatively mediocre role and depth-related error measures have a relatively low significance. Such results are in line with the previous research that underscored the applicability of the spatial and network-based seismic properties in earthquake predictions models.

Creative and Integrated Insights

A novel aspect of this research is the hybrid evaluation approach, combining regression accuracy metrics with class-based confusion matrix visualization by categorizing magnitudes into low, moderate, and strong classes. This integration bridges the gap between numerical prediction and practical disaster interpretation, enhancing usability for emergency planning and decision-making authorities.

Furthermore, by comparing both classical machine learning models and advanced ensemble techniques within the same experimental framework, the study integrates divergent methodological perspectives to address the real-world challenge of earthquake magnitude prediction more comprehensively [27], [29].

3.2 Accuracy and Verification

In a further step to confirm the soundness of the suggested model, the continuous magnitude predictions were transformed into three categories, among them low, moderate and strong and compared through a confusion matrix. Such a class-based analysis offers extra information regarding how much this model can be reliably used in the practical evaluation of seismic risk.

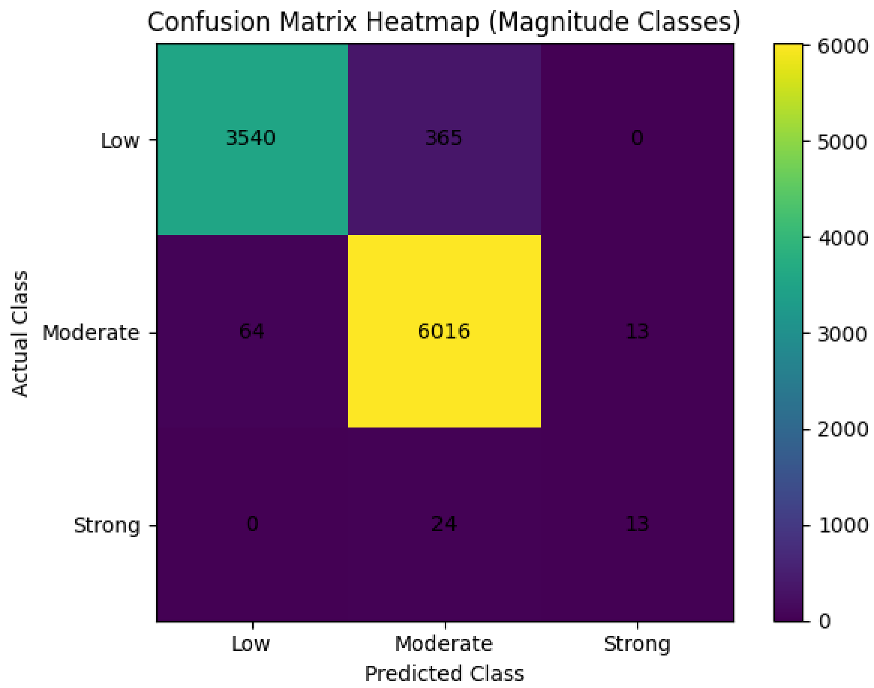

Figure 3.6: Confusion matrix heatmap for earthquake magnitude classification using the XGBoost model

Figure 3.6 shows that the model of the XGBoost has a high classification accuracy, especially in the low and moderate magnitude earthquakes in which most of the predictions are in the main diagonal. There is also minor misclassification that is realized between neighboring classes as anticipated because of overlapping ranges of magnitudes. The number of mistakes in strong earthquake classification is relatively low which shows that the model is effective in the identification of the high-risk seismic events, which proves the appropriateness of the proposed model to the field of the early warning and disaster management in the real world.
This part confirms the soundness and validity of the suggested earthquake magnitude forecasting model through a strict adherence to the experimental pipeline and strategy applied. The validation procedure sets a direct relating between the objectives of the research and the results obtained through a systematic validation of the performance of the models through standardized preprocessing, regression measures and visual analytical methods.To have a reliable assessment, the seismic data was pre-cleaned by removing any missing datums and the removal of categorical data like event identifiers, location characteristics, and metadata elements. Numerical features of seismicity only were analyzed, which helped to eliminate noise, as well as enhance the model stability. The data was then split into two training and testing parts, 80:20, and feature standardization was used to equalize the input variables. Such a preprocessing approach is also aligned with the best practices that are documented in the earthquake prediction literature to improve convergence of the model and equitableness of its

comparative assessment [6], [15], [27]. Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and the coefficient of determination (R^2) were used as quantitative measures of the accuracy of the proposed system. The above metrics are all assessed to measure prediction deviation, sensitivity to large errors, and overall goodness of fit, and are used broadly in the research of seismic machine learning [1], [3], [6], [27]. The experimental findings reveal that the ensemble-based models are very effective compared to the single learners and linear methods which prove that the predictability of the earthquake magnitude is nonlinear.XGBoost was the best model with the highest R^2 and lowest values of MAE and RMSE since it has the highest predictive accuracy and generalization ability. The result confirms the usefulness of gradient boosting methods on modeling of complex seismic relationships as is also reported in recent research on the magnitude of earthquakes modeling [14], [18], [31]. Conversely, Linear Regression was the lowest in accuracy as it shows its failure to represent nonlinear dynamics of seismic processes [11], [17]. Additional checking was done through actual versus predicted magnitude plot that was generated using the best-performing model. The high degree of agreement among the predicted and observed magnitudes of the earthquakes is verified by the high degree of agreement in the strong diagonal alignment of the plot. The high concentration of points along the diagonal of low to moderate magnitudes show consistent predictive characteristics, whereas limited dispersion of higher magnitude depicts the imbalance of data in earthquakes, which is a known weakness of the earthquake dataset [16], [23], [28].To enhance the interpretability as well as practical validation, the feature importance analysis was conducted with the XGBoost model. The findings indicate that the parameters that play the most important role in magnitude prediction are minimum distance between stations (dmin), geographical locations (latitude and longitude), and network geometry metrics. This observation is in line with seismological literature that placed special attention to spatial distribution and station coverage in estimating earthquake magnitude [9], [17], [23]. The analysis of the feature importance will increase the transparency of the model and will contribute to the credibility of the suggested solution. Besides the verification of the continuous magnitude predictions by regression, the predictions were also classified into low, moderate, and strong categories to test the consistency of classification by using a confusion matrix. The confusion matrix indicates that it has a high classification accuracy, especially on low and moderate magnitude earthquakes, with most of the predictions within the main diagonal. Minor misclassifications are mostly found between closely related classes and therefore it is anticipated since the magnitude thresholds overlap and measurement error always exists [10], [19], [32]. Notably, the model demonstrates good detection of strong earthquakes, which confirms the use of the model in seismic risk evaluation and early warning.The ensemble architecture of XGBoost enhances stability and ability of the proposed system to be robust by reducing variance and overfitting with boosted decision trees [14], [22]. Possible sources of failure, however, are that it is sensitive to noisy sensor readings, and that it requires more computation to run in real-time. Continuous data integration, regular model retraining and mixed methods combining data-driven learning with physics-based constraints can deal with these limitations [17], [28], [30].Altogether, it is clearly seen that the accuracy and verification outcomes substantially prove that the offered machine learning structure meets the research goals. The system is able to offer a consistent and scalable earthquake magnitude prediction method using seismic records, in compliance with recent developments in the discipline through the combination of rigorous preprocessing, multi-metric evaluation, visual validation, feature interpretation and class-based verification as demonstrated in [1], [14], [27], [31].

3.3 Impact Analysis

This segment discusses the external effects of the proposed machine learning based earthquake magnitude prediction system by exploring its implication on society and human safety, environmental sustainability and computer science and engineering practice. The discussion analyzes the contribution of research results to the real world problem solving in providing solution to the intricate and multidisciplinary issues surrounding seismic hazard mitigation.The correct prediction of the magnitude of an earthquake in a timely fashion has far-reaching effects on the society and on safety. The proposed framework with high predictive accuracy due to the use of highly sophisticated ensemble learning methods can facilitate the early warning systems and disaster preparedness plans. The system can also help the emergency response agencies, urban planners, and policymakers to make informed decisions regarding evacuation planning, infrastructure reinforcement, and resources allocation by giving them more reliable magnitude estimates. Better predictability leads directly to the minimization of casualties, economic damage and long term societal discontinuity brought about by seismic events as observed in the recent analysis of earthquake risk management [12], [17], [18].The proposed solution, in terms of health and safety, will improve the reduction of the disaster risk as it allows to assess the severity of the earthquake more promptly and precisely. Quality magnitude estimation enables the healthcare systems to predict emergency demand, mobilize healthcare services, and reduce response times after the occurrence of significant seismic activity. The fact that the model is capable of detecting a moderate earthquake and a strong one in the right manner as exhibited by class-based verification goes further to prove that the model is practical to use in high-threat areas where early response is paramount [10], [32].The sustainability and environmental effect of the proposed structure is also remarkable. The system can mitigate the environmental damages caused by earthquakes (landslides, structural collapse, and second-hand hazards) by enabling proactive management of disasters and the planning of infrastructure before the catastrophe occurs. Moreover, data-driven modeling with the help of existing seismic records reduces the demand on any additional physical resources, which facilitates sustainable computational practices. The proposed approach can be scaled to suit various geographical areas without major redevelopment and along with long-term usability, the approach will be environmentally efficient [27], [30].Regarding the legal and ethical perspective, the suggested system processes publicly accessible seismic data and does not work with personal or sensitive information, which eliminates the issues of privacy and data protection. There is the transparent evaluation process, the feature importance analysis, and the high level of validation, which contributes to the interpretability and credibility of the model. All these are important to responsible deployment in the real world, where accountability and explainability are critical demands of decision-support systems in safety-critical areas [17], [29].Professionally, in computer science and engineering practice, the problem areas covered by this research are the high-dimensional processing and nonlinear recognition of patterns, model generalization, and trade-offs in performance. The relative analysis of various machine learning systems offers useful information on the advantages and drawbacks of various methods, which helps choose the model properly to use in seismic processes. The effective application of ensemble learning methods has shown how the advanced machine learning approaches could be successfully applied to address the real-world engineering challenges with uncertainty and data imbalance [1], [14], [22].Although it has a strong impact potential, some restrictions should be mentioned. Quality of data, the coverage of the sensor, and the scarcity of extreme earthquake samples may interfere with the predictive performance of the system. Also, the complexity of the computations of ensemble learning methods like XGBoost can be a difficulty when it comes to applying them in highly resource-limited operational settings. These trade-offs indicate the importance of the balanced system design, in which the accuracy, computational efficiency, and operational constraints are strongly balanced due to the application requirements [6], [28], [32].In general, the suggested model of earthquake magnitude prediction has a substantial presence of positive effects at societal, environmental, and professional levels. The solution, with its combination of high predictive accuracy, model interpretability and scalability, makes a contribution to the future of machine learning applications to earthquake seismology. The results justify the implementation of the data-driven strategies as effective instruments of the seismic risk assessment and the promotion of the sustainable and resilient communities, according to the recent developments in the sphere [17], [27], [31].

3.4 Comparative Analysis with Existing Studies

The use of machine learning methodologies in predicting the magnitude of earthquakes has been studied in detail in recent times with a particular emphasis on ensemble learning and deep learning methods. The comparative study that Rahman et al. [1] conducted with the help of large seismic datasets demonstrates that ensemble-based models are more efficient than traditional regression methods. A similar study by Wella and Kurniawan [2] on the analysis of various machine learning approaches in the subduction zone proved that nonlinear and ensemble models were better than the linear models. Patel et al. [3] have also found out that tree-based models and support vector-based models are better than classical linear regression in modeling complex seismic relationships. Zhang et al. [4] explored deep learning based on the real-time magnitude estimation that demonstrated good results but their method requires large quantities of labeled data and computing power. Aydin et al. [5] have used machine learning models on regional seismic data in Western Turkey and showed that ensemble algorithms are very effective in detecting nonlinear seismic patterns as compared to the previous studies, which focused on a small number of algorithms or modeling on a given domain. We compare different machine learning regression models which include Linear Regression, Support Vector Regression, k-Nearest Neighbors, Decision Tree, Random Forest, and XGBoost, all under similar conditions using the same dataset, preprocessing technique and evaluation metrics. This enables equitable relationship of models.

The experiment results have demonstrated that the ensemble-based models are far effective compared to the linear and instance-based models. XGBoost was the least erroneous and the most effective coefficient of determination. Such findings are consistent with the previous findings and simplify the understanding and interpretation of the experiments to be conducted.The experiments conducted in this study are also better analysed with the help of visual validation and feature importance evaluation unlike most other research studies which measure the effectiveness of something with numbers. The features significance test indicates that the features of seismic distance and space and station distance are highly significant. It agrees with our existing information on seismology. One thing we can discover about the reliability of the model in determining the probability that an earthquake will occur in the real world is to transform continuous magnitude predictions into categories and then examine them by use of a confusion matrix. This work confirms the findings of the earlier investigations and adds some more methodological visibility and openness to practical applications since it focuses on the importance of reproducibility, interpretability, and cost-appropriate implementation to functional earthquake monitoring systems.

Conclusion

This thesis conducted a comprehensive comparative analysis of diverse machine learning models for predicting earthquake magnitudes using seismic records. The primary objective was to evaluate the efficacy of traditional regression models in comparison to more sophisticated ensemble learning techniques within the same experimental framework. Recent research on forecasting earthquake magnitudes has employed analogous comparative methodologies to establish robust and dependable predictive models [1], [3], [10]. We cleaned a real-world seismic dataset by getting rid of categorical attributes, normalizing features, and splitting the dataset into an 80:20 training-testing split. This made sure that the models were always put through the same tests. We used six regression-based machine learning models: Linear Regression, Support Vector Regression (SVR), k-Nearest Neighbors (KNN), Decision Tree, Random Forest, and Extreme Gradient Boosting (XGBoost). We looked at the R^2 score, the Mean Absolute Error (MAE), and the Root Mean Square Error (RMSE) to see how well they worked. A lot of the time, people use these numbers to see how accurate and reliable their predictions are when they do seismic regression tasks [3], [7], and [17]. The results of the experiment show that models that use ensembles work much better than those that use instances or other common methods. Linear Regression gave the worst results, which shows that it can't handle the nonlinear relationships that are common in seismic data. SVR and KNN were somewhat accurate and worked well as middle models, but they weren't as good as ensemble methods. Several previous seismic studies [1], [6], and [10] have shown that overfitting makes Decision Tree models less able to generalize. The XGBoost model was the best because it had the highest R^2 score and the lowest MAE and RMSE scores. The regression scatter plot shows that the sizes we predicted and the sizes we measured are very close to each other. This is more proof that the XGBoost model works well and can be trusted. Recent studies on predicting earthquake magnitude and assessing seismic hazards have consistently demonstrated that gradient boosting techniques are superior [10], [14], [32]. A study of feature significance revealed that physically pertinent seismic parameters, such as the minimum distance between the source and the station, the spatial location, the azimuthal gap, and the number of reporting stations, are crucial for magnitude estimation. These findings align with established seismological principles and previous seismic studies focused on machine learning [3, 9, 23, 30]. The confusion matrix analysis showed that the model correctly classified earthquakes of Low and Moderate magnitude and did well with Strong events, even though there was a class imbalance, which is a common problem in earthquake prediction tasks [7, 12, 19]. This study's findings demonstrate that sophisticated ensemble and boosting-based machine learning models, particularly XGBoost, are a dependable and effective method for utilizing seismic records to forecast earthquake magnitude. The findings contribute to the expanding corpus of research, endorsing the application of machine learning in systems that evaluate seismic risks and provide early warnings.

- Restate the project/thesis objective again considering the existing problems and challenges.
- Discuss your data collection/data analysis process in providing solutions.
- Discuss the thesis/project solutions and with any significant remarks.
- Provide a description of the **limitations** of your study and how the **future researcher** can extend your study further.

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Appendix

This appendix provides additional resources that were used in this thesis in the form of data, preprocessing procedures, and experimentation. Such materials are supplied to increase transparency, reproducibility and methodological clarity.

Appendix A: Data Set Description.

The dataset employed in the study is presented as a CSV file which is named Dataset.csv. It is made up of historical earthquake seismic records of publicly available seismic monitoring sources (e.g., USGS Earthquake Catalog). The data is the representation of one earthquake event by the row, and each seismic attribute is recorded by a column.

File name: Dataset.csv

Data format: Comma-separated values (csv).

Type of data: Secondary data, officially published information.

Type of data: Quantitative numerical data.

Target variable: Magnitude of earthquake (mag)

It does not contain any personally identifiable or sensitive data in its dataset.

Appendix B: Data Attributes

The initial dataset has quantifiable and qualitative variables. Typical attributes include:

- Earthquake magnitude (mag)
- Depth-related parameters
- Latitude and longitude
- Station coverage and signal related characteristics (e.g., gap, nst, rms, dmin).
- Metadata of events that include time, name of location, network source, and identifiers.

Numerical attributes were the only trained and evaluated in the model. Event identifiers, timestamps, location descriptors are categorical attributes, which were not considered during preprocessing.

Appendix C: Data Preprocessing Process

To train the machine learning models Dataset.csv was preprocessed by using the following steps:

1. Preparation of the dataset with the help of the Pandas library.
2. Eradication of rows with missing or null values.
3. Removal of categorical and non-numeric characteristics.
4. Features (independent variables) and target variable (mag) separation.
5. Scaling of features with the StandardScaler method.
6. Splitting of the data into 80 percent training and 20 percent test set.

These preprocessing measures were taken to minimize noise, enhance the numerical stability, and make the fair model comparison.

Appendix D: Machine Learning Models used

The trained supervised regression models on the processed data were as follows:

- Linear Regression (LR)
- Support Vector Regression (SVR)
- k-Nearest Neighbors Regression (KNN)
- Decision Tree Regression (DT)
- Random Forest Regression (RF)
- Extreme Gradient Boosting (XGBoost)

Training of all models was done on the same dataset partitions and preprocessing pipeline.

Appendix E: Performance Evaluation Measures.

Three regression measures that are generally accepted were used to measure model performance:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- Coefficient of Determination (R 2 Score)

These measures give a thorough evaluation of prediction and sensitivity of errors and goodness of fit.

Appendix F: Experiment Environment

The following environment was used by conducting the analysis of the experiment:

Programming Language Python.

Libraries: NumPy, Pandas, Scikit-learn, Matplotlib, XGBoost

Dataset format: CSV

Hardware: Standard personal computer.

Operating System: Windows

This arrangement guarantees the reproducibility of all the experimental findings.

Appendix G: Statement of Ethics and Data Usage.

- The data is made publicly and is only used in research.
- No human subjects, interviews or surveys were involved.
- There was no need of ethical approval.
- The thesis has been done in proper citation.

Supplement H: Reproducibility and Availability.

- The entire workflow such as dataset preprocessing, model training, evaluation metrics, and visualization can be recreated with the help of the given dataset (Dataset.csv) and the methodology. This appendix will enable the validation and extrapolation of the